

## NEXT.JS + GITLAB CI/CD DEPLOYMENT TO VPS

A complete, beginner-friendly, production-ready guide.

This guide explains exactly how to deploy a Next.js project to a VPS using GitLab CI/CD – using the simple

### 1. PREPARE YOUR VPS

=====

SSH into your VPS:

```
ssh root@YOUR_SERVER_IP
```

Install NVM + Node.js:

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/master/install.sh | bash
source ~/.bashrc
nvm install 20
node -v
npm -v
```

Install pnpm + pm2 globally:

```
npm install -g pnpm pm2
```

Check paths:

```
which pnpm
which pm2
```

### 2. CLONE YOUR PROJECT ON VPS

=====

```
cd /root
mkdir devkon-server
cd devkon-server
git clone https://gitlab.com/your-group/sf-lp-fashion-1.git
```

### 3. CHANGE REMOTE TO SSH

=====

```
cd /root/devkon-server/sf-lp-fashion-1
git remote set-url origin git@gitlab.com:your-group/sf-lp-fashion-1.git
git remote -v
```

### 4. VPS → GITLAB SSH DEPLOY KEY

=====

Generate SSH keypair on VPS:

```
ssh-keygen -t ed25519 -f /root/.ssh/gitlab_sf_lp -C "vps-ci-deploy"
```

Add /root/.ssh/gitlab\_sf\_lp.pub to GitLab Deploy Keys.

SSH Config:

```
cat > /root/.ssh/config <<EOF
Host gitlab.com
  HostName gitlab.com
  User git
  IdentityFile /root/.ssh/gitlab_sf_lp
  StrictHostKeyChecking accept-new
EOF
```

Test:

```
git pull origin dev
```

### 5. GITLAB CI → VPS SSH KEY (HARDCODED)

=====

Extract your local private key:

```
cat ~/.ssh/id_ed25519
```

## 6. CREATE .gitlab-ci.yml

=====

```
stages:
  - deploy

deploy_to_vps:
  stage: deploy
  image: alpine:latest
  only:
    - dev
  before_script:
    - apk add --no-cache openssh-client
    - mkdir -p ~/.ssh
    - |
      cat > ~/.ssh/id_ed25519 << 'EOF'
      -----BEGIN OPENSSH PRIVATE KEY-----
      YOUR_PRIVATE_KEY_HERE
      -----END OPENSSH PRIVATE KEY-----
      EOF
    - chmod 600 ~/.ssh/id_ed25519
  script:
    - echo "Connecting to VPS..."
    - >
      ssh -i ~/.ssh/id_ed25519 -o StrictHostKeyChecking=no root@YOUR_SERVER_IP
      "bash -lc 'export PATH=$PATH:/root/.nvm/versions/node/v20.x.x/bin &&
      cd /root/devkon-server/sf-lp-fashion-1 &&
      git pull origin dev &&
      pnpm install &&
      pnpm build &&
      pm2 restart storefront-landing-page'"
```

## 7. START PM2 APP ON VPS

=====

```
cd /root/devkon-server/sf-lp-fashion-1
pnpm install
pnpm build
pm2 start "pnpm start" --name storefront-landing-page
pm2 save
pm2 startup
```

## 8. RUN DEPLOYMENT

=====

```
git add .
git commit -m "Enable CI/CD"
git push origin dev
```

## DEPLOYMENT FLOW

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1. Push code
2. GitLab CI starts
3. CI SSH into VPS
4. VPS runs git pull
5. pnpm install
6. pnpm build
7. PM2 restart
8. Deployment complete